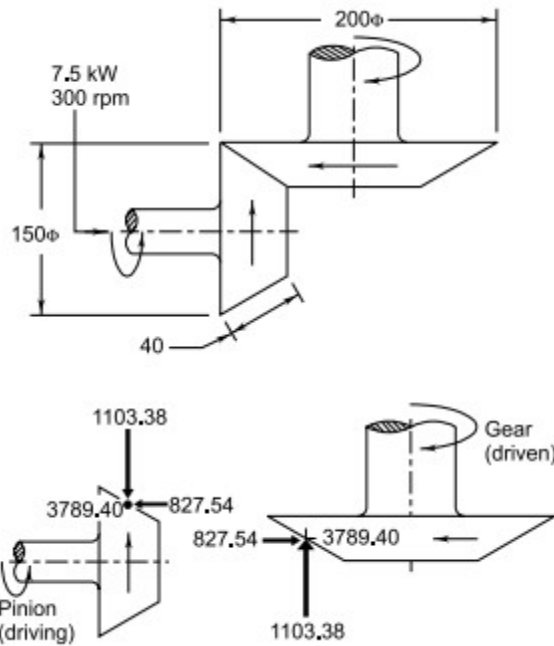


Instructions:

1. All questions are compulsory.
2. Use of non-programmable scientific calculator is allowed.
3. Assume suitable data whenever necessary and mention it clearly.

	Level/CO	Marks
Q.1 Solve the following.		
A) Give the advantages and disadvantages of hydrostatic and hydrodynamic bearings.	Apply/CO1	6
B) A single-row deep groove ball bearing is subjected to a radial force of 8 kN and a thrust force of 3 kN. The shaft rotates at 1200 rpm. The expected life L_{10h} of the bearing is 20 000 h. The minimum acceptable diameter of the shaft is 75 mm. Select a suitable ball bearing for this application.	Apply/CO2	6
Q.2 Solve any one.	Apply/CO5	12
A) A pair of spur gears with 20° full-depth involute teeth consists of a 19 teeth pinion meshing with a 40 teeth gear. The pinion is mounted on a crankshaft of 7.5 kW single cylinder diesel engine running at 1500 rpm. The driven shaft is connected to a two-stage compressor. Assume the service factor as 1.5. The pinion as well as the gear is made of steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$). The module and face width of the gears are 4 and 40 mm respectively. (i) Using the velocity factor to account for the dynamic load, determine the factor of safety. (ii) If the factor of safety is two for pitting failure, recommend surface hardness for the gears. (iii) If the gears are machined to meet the specifications of Grade 8, determine the factor of safety for bending using Buckingham's equation for dynamic load. (iv) Is the gear design satisfactory? If not, what is the method to satisfy the design conditions? How will you modify the design?		
B) A pair of parallel helical gears consists of 24 teeth pinion rotating at 5000 rpm and supplying 2.5 kW power to a gear. The speed reduction is 4 : 1. The normal pressure angle and helix angle are 20° and 23° respectively. Both gears are made of hardened steel ($S_{ut} = 750 \text{ N/mm}^2$). The service factor and the factor of safety are 1.5 and 2 respectively. The gears are finished to meet the accuracy of Grade 4. (i) In the initial stages of gear design, assume that the velocity factor accounts for the dynamic load and that the face width is ten times the normal module. Assuming the pitch line velocity to be 10 m/s, estimate the normal module. (ii) Select the first preference value of the normal module and calculate the main dimensions of the gears. (iii) Determine the dynamic load using Buckingham's equation and find out the effective load for the above dimensions. What is the correct factor of safety for bending? (iv) Specify surface hardness for the gears, assuming a factor of safety of 2 for wear consideration.		
Q.3 Solve the following.		
A) A pair of bevel gears transmitting 7.5 kW at 300 rpm is shown in Figure. The pressure angle is 20° . Determine the components of the resultant gear tooth force and draw a free-body diagram of forces acting on the pinion and the gear.	Remember/CO4	6



- B) A pair of worm gears is designated as, 1/30/10/8. Calculate (i) the centre distance; (ii) the speed reduction; (iii) the dimensions of the worm; and (iv) the dimensions of the worm wheel

Apply/CO4

6

Q.4 Solve any two.

- A) The turning moment diagram of a multi-cylinder engine is drawn with a scale of (1 mm = 1°) on the abscissa and (1 mm = 250 N-m) on the ordinate. The intercepted areas between the torque developed by the engine and the mean resisting torque of the machine, taken in order from one end are – 350, + 800, – 600, + 900, – 550, + 450 and – 650 mm². The engine is running at a mean speed of 750 rpm and the coefficient of speed fluctuations is limited to 0.02. A rimmed flywheel made of grey cast iron FG 200 ($\rho = 7100 \text{ kg/m}^3$) is provided. The spokes, hub and shaft are assumed to contribute 10% of the required moment of inertia. The rim has rectangular cross-section and the ratio of width to thickness is 1.5. Determine the dimensions of the rim.
- B) Explain the selection of flat-belts from manufacturer's catalogue.
- C) Explain the selection of V-belts from manufacturer's catalogue.

Apply/CO5

6

Remember/CO5

6

Remember/CO5

6

Q.5 Solve any two.

- A) An oil immersed multi-disk clutch, with moulded asbestos on one side and steel disks on the other, is used in an application. The torque transmitted by the clutch is 75 N-m. The coefficient of friction between the asbestos lining and the steel plate in the wet condition is 0.1. The permissible intensity of pressure on the asbestos lining is 500 kPa. The outer diameter of the friction lining is kept as 100 mm due to space limitation. Assuming uniform wear theory, calculate the inside diameter of the disks, the required number of disks and the clamping force.
- B) A cone clutch with asbestos friction lining transmits 30 kW power at 500 rpm. The coefficient of friction is 0.2 and the permissible intensity of pressure is 0.35 N/mm². The semi-cone angle α is 12.5°. The outer diameter is fixed as 300 mm from space limitations. Assuming uniform wear theory, calculate: (i) the inner diameter; (ii) the face width of the friction lining; and (iii) the force required to engage the clutch.
- C) A differential band brake is shown in Figure. The width and the thickness of the steel band are 100 mm and 3 mm respectively and the maximum tensile stress in the band is 50 N/mm². The coefficient of friction between the friction lining and the brake drum is 0.25. Calculate: (i) the tensions in the band; (ii) the actuating force; and (iii) the torque capacity of the brake. Find out whether the brake is self-locking.

Apply/CO3

6

Apply/CO3

6

Apply/CO3

6

Table: Dimensions and static and dynamic load capacities of single-row deep groove ball bearings

Principal dimensions (mm)			Basic load ratings (N)		Designation
<i>d</i>	<i>D</i>	<i>B</i>	<i>C</i>	<i>C₀</i>	
70	90	10	12100	9150	61814
	110	13	28100	19000	16014
	110	20	37700	24500	6014
	125	24	61800	37500	6214
	150	35	104000	63000	6314
75	180	42	143000	104000	6414
	95	10	12500	9800	61815
	115	13	28600	20000	10615
	115	20	39700	26000	6015
	130	25	66300	40500	6215
	160	37	112000	72000	6315
	190	45	153000	114000	6415

Table: Values of the Lewis form factor Y for 20° full-depth involute system

<i>z</i>	<i>Y</i>	<i>z</i>	<i>Y</i>	<i>z</i>	<i>Y</i>
15	0.289	27	0.348	55	0.415
16	0.295	28	0.352	60	0.421
17	0.302	29	0.355	65	0.425
18	0.308	30	0.358	70	0.429
19	0.314	32	0.364	75	0.433
20	0.320	33	0.367	80	0.436
21	0.326	35	0.373	90	0.442
22	0.330	37	0.380	100	0.446
23	0.333	39	0.386	150	0.458
24	0.337	40	0.389	200	0.463
25	0.340	45	0.399	300	0.471
26	0.344	50	0.408	Rack	0.484

Table: Tolerances on the adjacent pitch

Grade	<i>e</i> (microns)
1	0.80 + 0.06 ϕ
2	1.25 + 0.10 ϕ
3	2.00 + 0.16 ϕ
4	3.20 + 0.25 ϕ
5	5.00 + 0.40 ϕ
6	8.00 + 0.63 ϕ
7	11.00 + 0.90 ϕ
8	16.00 + 1.25 ϕ
9	22.00 + 1.80 ϕ
10	32.00 + 2.50 ϕ
11	45.00 + 3.55 ϕ
12	63.00 + 5.00 ϕ

Table: Recommended series of module (mm)

Choice-1	1.0	1.25	1.5	2.0	2.5	3.0	4.0
Preferred	5.0	6.0	8.0	10	12	16	20
Choice-2	1.125	1.375	1.75	2.25	2.75	3.5	4.5
	5.5	7	9	11	14	18	

Table: Values of deformation factor C (N/mm²)

<i>Materials</i>		<i>14.5° full depth teeth</i>	<i>20° full depth teeth</i>	<i>20° stub teeth</i>
Pinion material	Gear material			
Grey CI	Grey CI	5500	5700	5900
Steel	Grey CI	7600	7900	8100
Steel	Steel	11 000	11 400	11 900